Week 5 | Data Visualisation: Design Choices Are Arguments

Reflection

This week I tried to turn the classification data from last week's group exercise into a visualisation, and it went badly at first. Our group had collected images of everyday objects and sorted them into categories like "food," "kitchen," "garden," and "other." I put the numbers into a simple bar chart in Google Sheets. It looked fine—clean, colourful, easy to read. But when I showed it to my group, someone pointed out that the "other" category was huge, and the chart made it look like a minor leftover. In reality, "other" was where all the ambiguous objects ended up—the plastic apple, the mug with the plant, the things that did not fit. By making "other" a small grey bar at the end, I had visually erased the most interesting part of our data: the fact that our categories failed. That was a real lesson. The chart was not neutral. It was telling a story I did not intend to tell.

This connects directly to Feigenbaum and Alamalhodaei's (2020) chapter on visual data storytelling. They argue that data visualisation is not just a way to present findings; it is a design process that shapes the message. Choices about colour, order, size, and grouping are not technical details—they are arguments. My bar chart made an argument that our classification worked, when actually it revealed that our categories were messy. Feigenbaum and

Alamalhodaei suggest that good data storytelling requires thinking about narrative before design. I did the opposite: I made the chart first and then noticed what story it was telling. If I had thought about the narrative first—"our categories broke down when objects did not fit"—I would have designed the visualisation differently, maybe showing the "other" category as the centre of the story rather than the margin.

The prompt asked about the ideological work of data visualisation. I think the ideological work is exactly this: visualisations make certain things visible and other things invisible, and they make those choices feel natural. A bar chart looks objective. It looks like the data is just "showing itself." But someone decided which categories to include, which colours to use, and what order to put them in. Those decisions carry values. D'Ignazio and Klein's (2020) feminist data visualisation principles are useful here. They argue that visualisations should challenge binaries, embrace pluralism, and make uncertainty visible. My chart did the opposite: it forced ambiguous objects into a binary (either "food" or "other"), and it hid uncertainty by making the "other" category look unimportant. A feminist approach would have made the ambiguity central, perhaps using a different visual form—like a network diagram or a Venn diagram—that shows overlap rather than forcing separation.

I also looked at one of The Pudding's data stories, "Who Gets Shipped and Why?" (Cai et al., 2024). What I liked was that the visualisations were not just

decoration; they were part of the argument. The colours and groupings changed as the story developed, and the visualisation made me feel the scale of the data rather than just reading a number. That is what Feigenbaum and Alamalhodaei mean by visual storytelling: the visualisation is not a final step after analysis; it is part of the analysis itself. What I disliked was how some charts were so stylised that I had to work hard to understand what the axes meant. That reminded me that "attractive" is not the same as "clear."

So what did this week teach me about digital methods and digital media? It taught me that data visualisation is a critical practice, not a technical one. The design process is where meaning gets made, and that means designers have responsibility. A visualisation can clarify or obscure, include or erase. As a digital media student, I need to ask not just "what does this chart show?" but "what does this chart hide, and why?" For my portfolio website, I have added a Week 5 page with my first bar chart, a revised version that makes the "other" category central, and a short note explaining why I changed it. I also included a screenshot of The Pudding story and a few notes on what worked and what did not.

Portfolio evidence: [insert link to your portfolio website Week 5 page here]

References

Cai, A., Sutanto, F., Diehm, J. and Ralph, C. (2024) 'Who gets shipped and why?', *The Pudding*, October. Available at: https://pudding.cool/2024/10/fanfic/

D'Ignazio, C. and Klein, L. F. (2020) *Data Feminism*. Cambridge, MA: The MIT Press.

Feigenbaum, A. and Alamalhodaei, A. (2020) 'Visual data storytelling', in *The Data Storytelling Workbook*. London: Routledge.